

A Hybrid 3D-2D Convolutional Neural Network with Spatial Attention for Enhanced Spectral-Spatial Classification of Hyperspectral Images

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Abstract—Hyperspectral image (HSI) classification using deep learning (DL) has proven to be highly effective. However, efficiently learning highly discriminative features from HSI is an open issue for the large dimensionality and complex spectral-spatial properties. The 2D Convolutional Neural Network (CNN) is an effective method for classifying Hyperspectral Images based on their spatial or spectral features. But HSI depends on both spectral and spatial classification. On the other hand, the 3D CNN is superior to the 2D CNN as it leverages both spectral and spatial features of the HSI. Nevertheless, in this case, computational complexity increases. This paper introduces a novel approach that utilizes a hybrid 3D-2D CNN structure with a spatial attention mechanism that specifically emphasizes spatial attributes. We utilize Z-score normalization followed by Incremental Principal Component Analysis (IPCA) for efficient dimensionality reduction and optimized memory usage on large datasets. Next, we use a spatial attention mechanism and a hybrid 3D-2D CNN to extract the features. In the hybrid CNN architecture, the 3D-2D branch merges the 3D CNN's capability to capture detailed spectral-spatial features with the 2D CNN's spatial refinement, further enhanced by a spatial attention mechanism. This approach balances comprehensive feature extraction with computational efficiency, improving accuracy and reducing overfitting. This results in great accuracy at a limited number of parameters. We have implemented the proposed hybrid CNN with attention mechanism on three hyperspectral datasets and it showcases exceptional performance in both accuracy and computational efficiency than other state-of-the-art methods.

Index Terms—Hyperspectral images (HSIs), CNN, 3D CNN, IPCA, Dimensionality Reduction, Spatial Attention, Z score normalization

I. INTRODUCTION

Hyperspectral imaging reveals a broad range of hidden features within the light spectrum, providing deeper insights into the composition and characteristics of materials and environments. This advanced imaging technique is vital in diverse fields such as astronomy, agriculture, and environmental monitoring. By capturing and analyzing light across numerous spectral bands, hyperspectral imaging allows for the precise identification of

various materials [1], enhancing our understanding of the world around us. This capability is particularly valuable in remote sensing, which has become a crucial aspect of modern technological advancements. Remote sensing, conducted through aircraft, satellites, or other platforms, offers critical information about our environment from a distance. Its applications span a wide array of fields, including geography, environmental research, and agriculture, and its utility extends from mapping land cover to aiding in disaster management and response efforts [2].

Within the domain of hyperspectral remote sensing, however, the process is complex and demanding due to the continuous spectral data acquired from multiple bands [3]. The vast amount of information obtained enables precise material characterization and significantly improves the ability to distinguish between different features. Yet, these advantages come with significant challenges. Hyperspectral image classification, for instance, faces the daunting task of managing high-dimensional data [4], which is further complicated by spectral variations and the limited availability of labeled datasets. Additionally, issues such as imbalanced class distributions and the computational demands of processing such large datasets exacerbate these challenges. Another critical aspect involves the effective integration of spatial and spectral data while minimizing noise and mitigating atmospheric effects, which are essential for accurate analysis and interpretation of hyperspectral images [4].

In recent years, deep learning (DL) has revolutionized the field of computer vision, largely due to its ability to autonomously extract intricate, high-level features from extensive datasets [5]. Among the various DL models, Convolutional Neural Networks (CNNs) have become prominent for their effectiveness in image processing and analysis. CNNs have become a key focus in HSIC because they efficiently model spatial and spectral dependencies in hyperspectral data [6].

Typically, CNN-based methods for HSIC involve extracting a small hyperspectral image (HSI) patch from the larger HSI dataset, which is then fed into the network. The CNN analyzes this patch to classify the central pixel, leveraging its deep layers to learn intricate patterns within the spectral and spatial dimensions. This patch-based approach allows CNNs to effectively handle the high-dimensional nature of hyperspectral data while capturing local contextual information, which is crucial for accurate classification. Moreover, advancements in CNN architectures, such as the integration of residual connections and attention mechanisms, have further enhanced their performance in HSIC [7].

This study presents an advanced Attention-Based Hybrid Convolutional Neural Network to improve spectral-spatial classification of hyperspectral images. By integrating traditional CNNs with attention mechanisms, the model highlights key spectral-spatial features and minimizes noise and irrelevant data. By integrating attention mechanisms within the CNN architecture, our approach not only captures important patterns in hyperspectral data but also significantly improves classification accuracy. This model's pioneering architecture proficiently manages the complexities of high-dimensional hyperspectral image (HSI) data, overcoming the shortcomings of traditional classification techniques.

II. LITERATURE REVIEW

Recent years have seen significant advancements in Convolutional Neural Networks (CNNs) techniques for Hyperspectral Image Classification (HSIC), particularly focusing on spatial features using 2D CNN models. [8] Typically, these spatial features are extracted independently, which limits the utilization of combined spatial and spectral information. To address this, a hybrid approach combining spectral CNN and spatial CNN has been introduced [9]. This approach leverages a 3D CNN to capture spectral-spatial features, followed by a 2D CNN to encapsulate higher-level abstract spatial information. Although this hybrid model surpasses conventional 3D models, it still encounters challenges in fully extracting intricate spatial abstractions.

Recently, Paoletti et al. [10] [11] introduced two pyramidal residual networks for hyperspectral image (HSI) classification, one focused on spectral information and the other utilizing both spectral and spatial data through a capsule network. Chen et al. [12] proposed a 3D CNN for feature extraction and classification, while Zhong et al. [13] presented a spatial-spectral residual network using identity-mapped residual blocks to link 3D convolution layers. Mou et al. [14] proposed an unsupervised approach leveraging residual CNNs. Muhammad et al. [15] introduced a 3D CNN model that divides the HSI cube into overlapping patches, generating 3D feature maps via a 3D kernel across contiguous spectral bands. Lu et al. [16] proposed a new band-grouping guided multi-attention module for spectral-spatial feature learning. Also, the Graph Attention Neural Network is implemented to enhance hyperspectral image classification by Rathakrishnan and Raja. [17]

Our literature review identified several shortcomings in the current state of research:

Despite the promise of CNNs for HSIC, their high memory demands and computational complexity hinder performance acceleration. This study applies CNNs to HSIC, aiming for high accuracy with controlled computational costs, particularly focusing on convergence time. To achieve this, we progressively modify a baseline model to maintain accuracy while reducing time complexity.

Preserving inter-channel relationships poses a significant challenge, as CNN models often operate on reshaped spectral bands or single-band data, which inadequately captures fine structural nuances. The pronounced class similarity, variability within classes, and overlapping regions in HSI data further complicate classification. To address these issues, the proposed architecture partitions the HSI cube into small, overlapping 3D patches, which are processed through a 3D-2D convolutional layer to generate feature maps. Concurrently, a parallel branch channels these features into a spatial attention block to preserve integrated spatial-spectral information. This method substantially enhances both classification precision and computational efficacy relative to current state-of-the-art techniques.

III. MATERIALS AND METHODS

A. Dataset

Salinas dataset was taken over Salinas Valley in California. To capture the image, the AVIRIS sensor was used, and the Salinas dataset (SD) provides comprehensive hyperspectral imagery. With dimensions of $512 \times 217 \times 224$ and a spatial resolution of 3.7 meters, SD includes 16 distinct classes, such as vineyards, crops, and bare soils [18].

The Indian Pines dataset (IPD) was acquired in northwestern Indiana with the AVIRIS sensor. It has a size of $145 \times 145 \times 224$ and spans a wavelength range of $0.4\text{--}2.5 \times 10^{-6}$ meters, encompassing diverse land cover types such as forest, agricultural fields, and other types of vegetation [19].

The Pavia University dataset (PUD) provides hyperspectral imagery with spatial dimensions of 610×610 and 103 spectral bands. Collected in Pavia, Italy using the ROSIS sensor, it has a spatial resolution of 1.3 meters and is classified into 9 ground truth categories.

B. Proposed Methodology

Let $X \in \mathbb{R}^{H \times W \times B}$ represent the hyperspectral image (HSI), where $H \times W$ denotes the spatial dimensions and B is the number of spectral bands. Hyperspectral images typically encompass data across multiple spectral bands, each exhibiting distinct scales and value ranges. To address this disparity, Z-score normalization is employed, transforming each band such that its mean becomes 0 and its standard deviation becomes 1, thereby eliminating scale differences and rendering the bands comparable. HSI pixels demonstrate high interclass similarity, intraclass variability, as well as overlapping and nested regions. To mitigate these challenges, Incremental Principal Component Analysis (iPCA) is applied to discard

redundant bands, reducing the number of spectral bands (B to L , where $B \gg L$) while preserving spatial integrity. The data preprocessing workflow is depicted in Fig. 1.

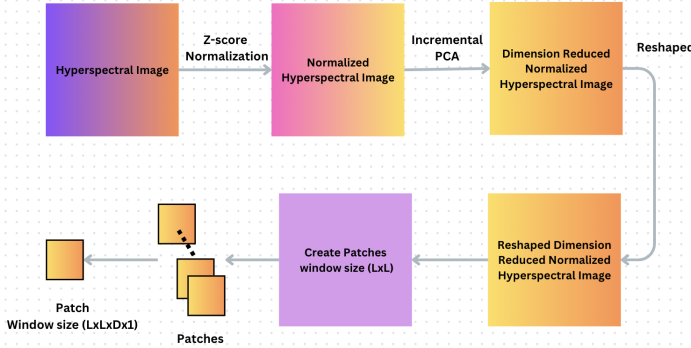


Fig. 1: Data Preprocessing

To enhance the classification of Hyperspectral Image (HSI) patches, we typically crop a patch $X_i \in \mathbb{R}^{s \times s \times L}$ from the larger image X , and input it into a Convolutional Neural Network (CNN) to extract both spectral and spatial features. Our proposed architecture involves two main branches within the network.

The first branch is dedicated to attention mechanism, focusing on extracting spatial attention features denoted as $O_{\text{Attention}}$. Meanwhile, the second branch employs a hybrid 3D-2D CNN architecture to capture spectral-spatial features, represented as O_{Conv} . The schematic overview of our proposed architecture is depicted in Figure 2.

The output features from both branches are then concatenated and fed into the output layer, which consists of a fully connected layer and then passed through a softmax function. The label vector y is computed by applying the softmax function to the output of the fully connected layer, which learns from the combined attention and convolutional features:

$$y = \text{Softmax}(f_{\text{FC}}(C(O_{\text{Attention}}, O_{\text{Conv}}))) \quad (1)$$

Here, $C(\cdot)$ represents the concatenation operation, $f_{\text{FC}}(\cdot)$ denotes the feature learning process by the fully connected layer, and y contains the classification probabilities for different categories.

C. Spatial Attention Branch

To generate spatial attention maps for each patch, our spatial attention branch aims to emphasize significant spatial information to aid feature learning and classification. To achieve this, we devised a learnable spatial attention block, illustrated in Figure 3. Initially, we apply downsampling to the input patch X to reduce redundancy and highlight important details. Instead of utilizing non-learnable pooling operations, we employ a learnable convolutional layer for downsampling. Specifically, we use a combination of 3D convolution and nearest interpolation, expressed as:

$$F_t = \text{upsampling}(W_a * X) \quad (2)$$

In this context, F_t symbolizes the feature maps that roughly capture prominent spatial information, while W_a signifies a 3D convolutional layer using a $3 \times 3 \times 1$ kernel size. To restore the feature maps to their original spatial dimensions, nearest-neighbor interpolation is employed in the $\text{upsampling}(\cdot)$ function.

Following this, F_t undergoes additional feature refinement through a 3D convolutional layer. The extracted features are then standardized via batch normalization and activated using the Mish function to produce attention weights. Ultimately, the attention maps F are computed through the following equation:

$$F = M(f_{\text{BN}}(W_b * F_t)) \quad (3)$$

Here, W_b corresponds to a 3D convolution using a $3 \times 3 \times 1$ kernel size, and M denotes the activation function (specifically, the Mish function is utilized here).

D. Convolutional Branch

The convolutional branch within the proposed architecture aims to capture spectral-spatial features from the hyperspectral image (HSI) patches. Leveraging a hybrid 3D-2D convolutional architecture, this branch operates on the input HSI data, employing successive Conv3D layers to process the three-dimensional nature of the spectral information. Each Conv3D layer is followed by batch normalization to stabilize training and the Mish activation function to introduce non-linearity. These convolutional layers enable the extraction of complex spectral features across multiple spectral bands while preserving spatial information. The architecture allows for flexibility in the number of convolutional layers and the choice of filter sizes, enabling adaptation to the intricacies of the HSI data. By systematically learning from the HSI patches, the convolutional branch facilitates the generation of informative feature representations, contributing to the subsequent classification task.

conv_layer1 consists of 8 filters, each having a kernel size of $3 \times 3 \times 7$. This leads to the application of 8 sets of 3D convolutional filters on the input hyperspectral picture. The *conv_layer2* function enhances the output of *conv_layer1* by utilizing 16 filters having a kernel size of $3 \times 3 \times 5$. This layer captures more complex and abstract features from the features obtained by *conv_layer1*. The tensor *reshaped_conv_layer2* preserves the spatial dimensions of the *conv_layer2* output while consolidating the depth information into a singular channel. *Conv_layer3* and *conv_layer4* are 2-D convolutional layers that are used on the altered output of *conv_layer2*. *Conv_layer3* consists of 32 filters, each with a kernel size of 3×3 . The reshaped output of *conv_layer2* undergoes 32 sets of 2D convolutional filters. The *conv_layer4* module applies 64 filters, which have a kernel size of 3×3 to the output of *conv_layer3*. The Mish activation function, denoted as *mish*,

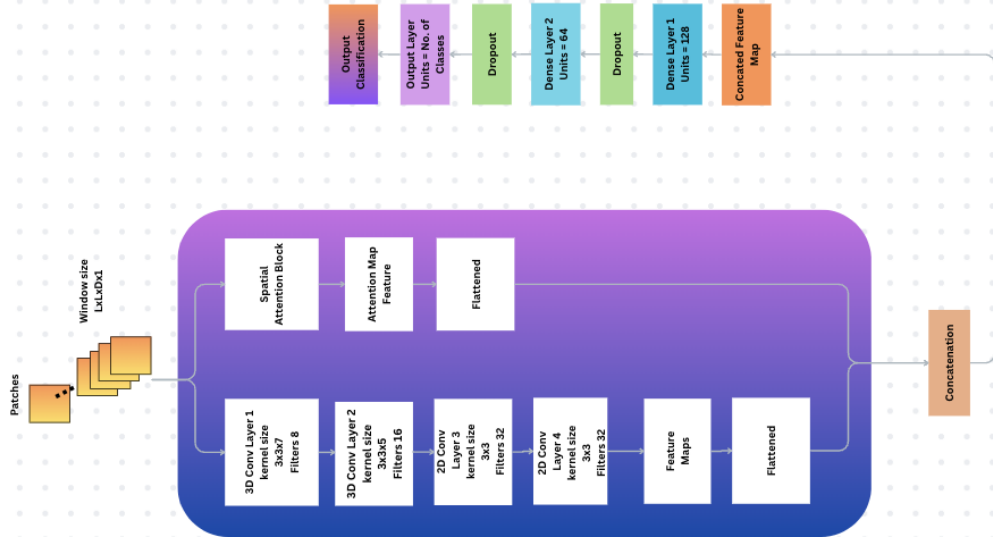


Fig. 2: Proposed Methodology

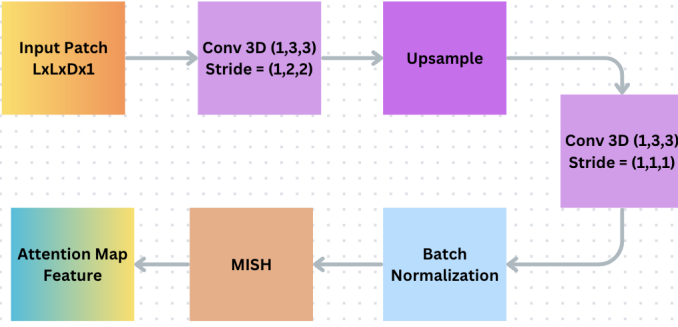


Fig. 3: Spatial Attention Block

is applied to all convolutional layers, specifically *conv_layer1*, *conv_layer2*, *conv_layer3*, and *conv_layer4*.

TABLE I: Total Number of Trainable Parameters for Different Models with a Patch Size of 11×11 .

Method	IP	PU	SA
iPCA+3DCNN	995,456	994,682	995,456
iPCA+Attention Map+2D-3DCNN+Modified Dense Layer	566,236	565,846	566,236

IV. EXPERIMENTAL RESULT ANALYSIS

A. Experimental Settings

Considering the size of the dataset, the model was configured to run on both CPU and GPU. And we run the code in Kaggle with accelerator GPU T4 x2. Building a dependable and accurate neural network that could make astute predictions using the given dataset was the primary goal of the training process.

B. Analysis and Outcomes

For all experiments, the learning rate is configured to 0.001. All layers use ReLU as the activation function, while the final layer uses softmax. The patch sizes are configured as $9 \times 9 \times 20$, $11 \times 11 \times 20$, $13 \times 13 \times 20$, $15 \times 15 \times 20$, $17 \times 17 \times 20$, $19 \times 19 \times 20$, $21 \times 21 \times 20$, $23 \times 23 \times 20$, and $25 \times 25 \times 20$. In each case, the 20 most informative spectral bands are selected via the iPCA method. From the experimental analysis, we have observed that the accuracy after 17×17 patch size is changing very slightly, that's why we have kept the analysis up to patch size 17×17 .

TABLE II: Comparison of Average Accuracy and Overall Accuracy for Different Patch Sizes

Patch Size	Average Accuracy (%)			Overall Accuracy (%)		
	IP	PU	SA	IP	PU	SA
9x9	98.11	96.11	98.91	99.00	96.42	99.11
11x11	99.15	99.20	99.17	99.78	99.81	99.77
13x13	99.00	99.13	99.00	99.43	99.71	99.61
15x15	99.11	99.14	99.20	99.63	99.76	99.84
17x17	99.20	99.22	99.24	99.80	99.84	99.80

From Table II, it can be concluded that a patch size of 11×11 is sufficient for the Pavia University, Salinas, and Indian Pines datasets, while patch sizes of 13×13 and 17×17 yield similar results.

For evaluation, the Kappa (κ) coefficient, overall accuracy (OA), and average accuracy (AA) are computed from the confusion matrices. The Kappa coefficient provides a statistical measure of agreement between predicted and actual classifications. Overall accuracy indicates the proportion of correctly classified samples out of the total test set, while average accuracy reflects the mean classification performance across all classes.

The accuracy metrics— κ , OA, and AA—based on the spatial dimensions handled by the proposed model are detailed in Tables III, IV, and V.

TABLE III: Comparing the proposed method with existing methods on Indian Pine Dataset (IP) (Where Patch size 11x11).

Model Name	AA	OA	Kappa
2D-CNN	68.32	80.27	75.26
3D-CNN	76.51	82.62	79.25
MS-3D-CNN	75.22	81.39	81.39
FC-3D-CNN	94.54	97.75	97.44
GATN-RTC	95.29	95.35	94.87
Proposed Methodology	99.15	99.78	98.99

TABLE IV: Comparing the proposed method with existing methods on Pavia University Dataset (PU) (Where Patch size 11x11).

Model Name	AA	OA	Kappa
2D-CNN	94.84	96.83	95.53
3D-CNN	97.03	96.34	94.90
MS-3D-CNN	97.52	95.95	93.40
FC-3D-CNN	97.89	98.40	97.89
GMA-net	98.84	98.78	98.39
GATN-RTC	96.27	96.15	95.98
Proposed Methodology	99.20	99.81	99.11

TABLE V: Comparing the proposed method with existing methods on Indian Salinas (SA) (Where Patch size 11x11).

Model Name	AA	OA	Kappa
2D-CNN	94.36	96.34	95.93
3D-CNN	89.63	85.00	83.20
MS-3D-CNN	96.66	94.20	93.61
FC-3D-CNN	98.80	98.06	97.85
GMA-net	98.57	96.46	96.06
GATN-RTC	94.62	95.37	95.64
Proposed Methodology	99.17	99.77	99.77

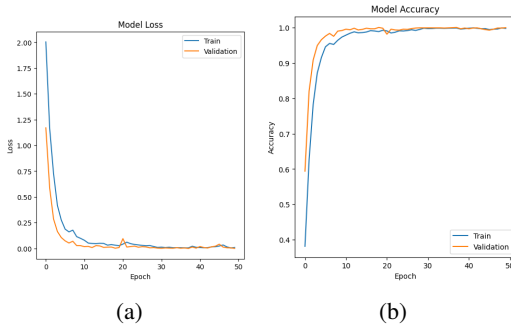


Fig. 4: (a) Training and Validation loss (IP) (b) Training and Validation Accuracy (IP)

In fig 4, 5, 6 shows the train validation loss and train validation accuracy for all 3 datasets. Where we can see both train and validation loss decrease over the epochs and the same goes for accuracy. So it can be said that there was no overfitting issue. And the model achieved higher accuracy on 50 epochs.

Regarding the comparison, the proposed model is rigorously evaluated against several advanced methodologies published

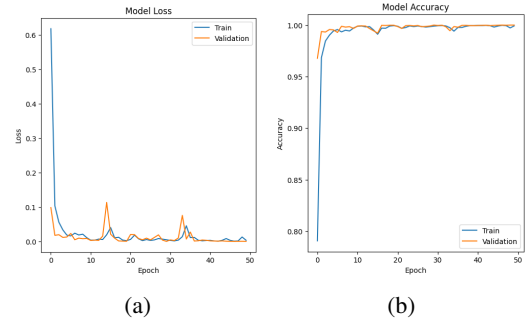


Fig. 5: (a) Training and Validation loss (PU) (b) Training and Validation Accuracy (PU)

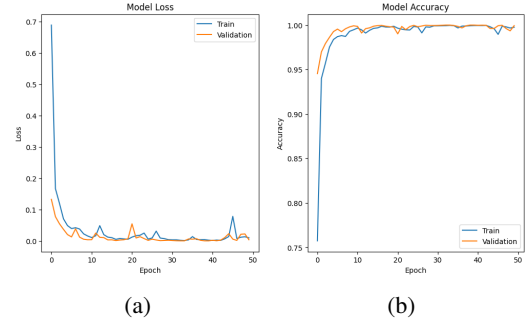


Fig. 6: (a) Training and Validation loss (SA) (b) Training and Validation Accuracy (SA)

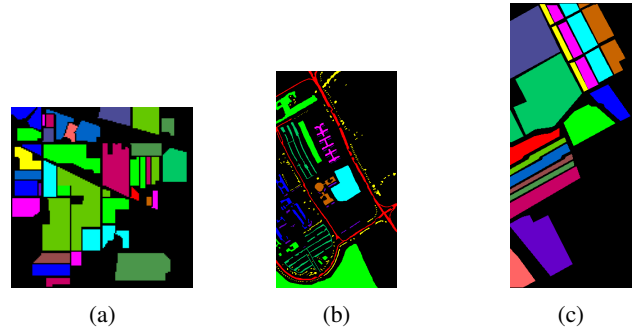


Fig. 7: Classification map image (a) Indian Pine (b) Pavia University and (c) Salinas Dataset.

in recent years. As evidenced by the experimental results in Tables III, IV, and V, our model not only demonstrates competitive performance but also outperforms other approaches in several instances. A meticulous comparative analysis of Fast and Compact 3D CNN (FC-3D-CNN), Multiscale 3D CNN (MS-3D-CNN), 3D-CNN/2D-CNN, Group Multi-Attention Network (GMA-net), Graph Attention Neural Network for Remote Target Classification (GATN-RTC), and our proposed architecture is conducted across diverse datasets, with all models being trained according to the specifications detailed in their respective publications. Our model's superior performance stems from the integration of Incremental PCA for more efficient dimensionality reduction, reducing computa-

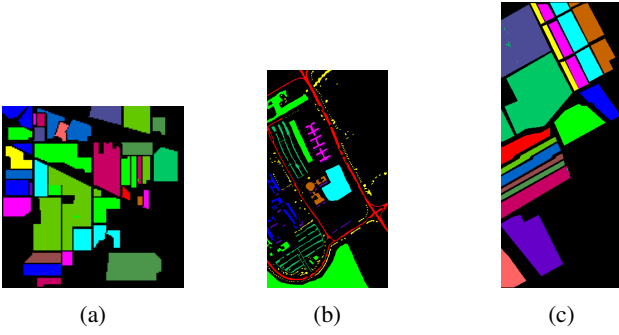


Fig. 8: Image by our model for (a) Indian Pine (b) Pavia University and (c) Salinas Dataset.

tional overhead, and the application of z-score normalization, which improved feature scaling and model convergence. The hybrid 3D-2D CNN architecture captured both spectral and spatial features more effectively than traditional 3D CNNs, while the spatial attention mechanism concentrated on crucial spatial regions, further boosting classification accuracy. As demonstrated in Tables III, IV, and V, the proposed method delivers significantly better results with fewer training samples, fewer convolutional layers, fewer filters, fewer epochs, and reduced computational time compared to state-of-the-art techniques.

V. CONCLUSIONS AND DISCUSSIONS

This paper proposed an advanced methodology for hyperspectral image classification, addressing the complexities associated with this challenging task. By integrating Incremental PCA (IPCA), and z-score normalization, our approach ensures efficient dimensionality reduction and uniform feature impact, overcoming issues related to data intricacy and limited labeled data. Our advocated hybrid 3D-2D Convolutional Neural Network (CNN) architecture, augmented with a self-attention mechanism, demonstrates enhanced feature extraction capabilities by capturing long-range dependencies in the hyperspectral data. Additionally, the adoption of the Mish activation function leads to improved performance in deep neural networks compared to traditional activation functions like ReLU. Through experimental validation using well-known hyperspectral datasets such as Indian Pines, Pavia University and Salinas, our methodology showcases exceptional results with significantly reduced parameter count. This translates to reduced processing requirements, offering a practical solution for accurate and efficient hyperspectral image analysis. In conclusion, our proposed methodology represents a significant advancement in hyperspectral image classification, addressing critical challenges and leveraging modern techniques to achieve high accuracy and reduced complexity. It holds promise for various applications requiring precise analysis of hyperspectral imagery. Future work could explore the application of transfer learning to enable the model to generalize more effectively across diverse hyperspectral datasets, potentially reducing training time. Additionally, enhancing the attention

mechanism by incorporating more advanced models, such as transformers, could further improve the ability to capture long-range dependencies.

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